

CLAIRE Design Documentation

Competency Learning Analytics for Insights and Reflection in Education

2025 – Jonas Van Hove (KU Leuven)

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Global Overview

CLAIRE is a Learning Analytics Dashboard (LAD) designed to support secondary school teachers during class councils. The tool transforms raw learning data into clear and actionable insights, helping teachers interpret student performance with a focus on competency-based learning.

CLAIRE was developed as a proof of concept (PoC) in collaboration with KU Leuven, GITO Overijse, and Eummena, as part of the master thesis project “Turning Learning Data into Meaningful Learning Analytics Dashboards” by Jonas Van Hove (2025).

The goal is to create a dashboard that is both intuitive and insightful, allowing teachers to identify trends, weaknesses, and strengths at a glance — facilitating timely and personalized interventions.

CLAIRE Dashboard Overview

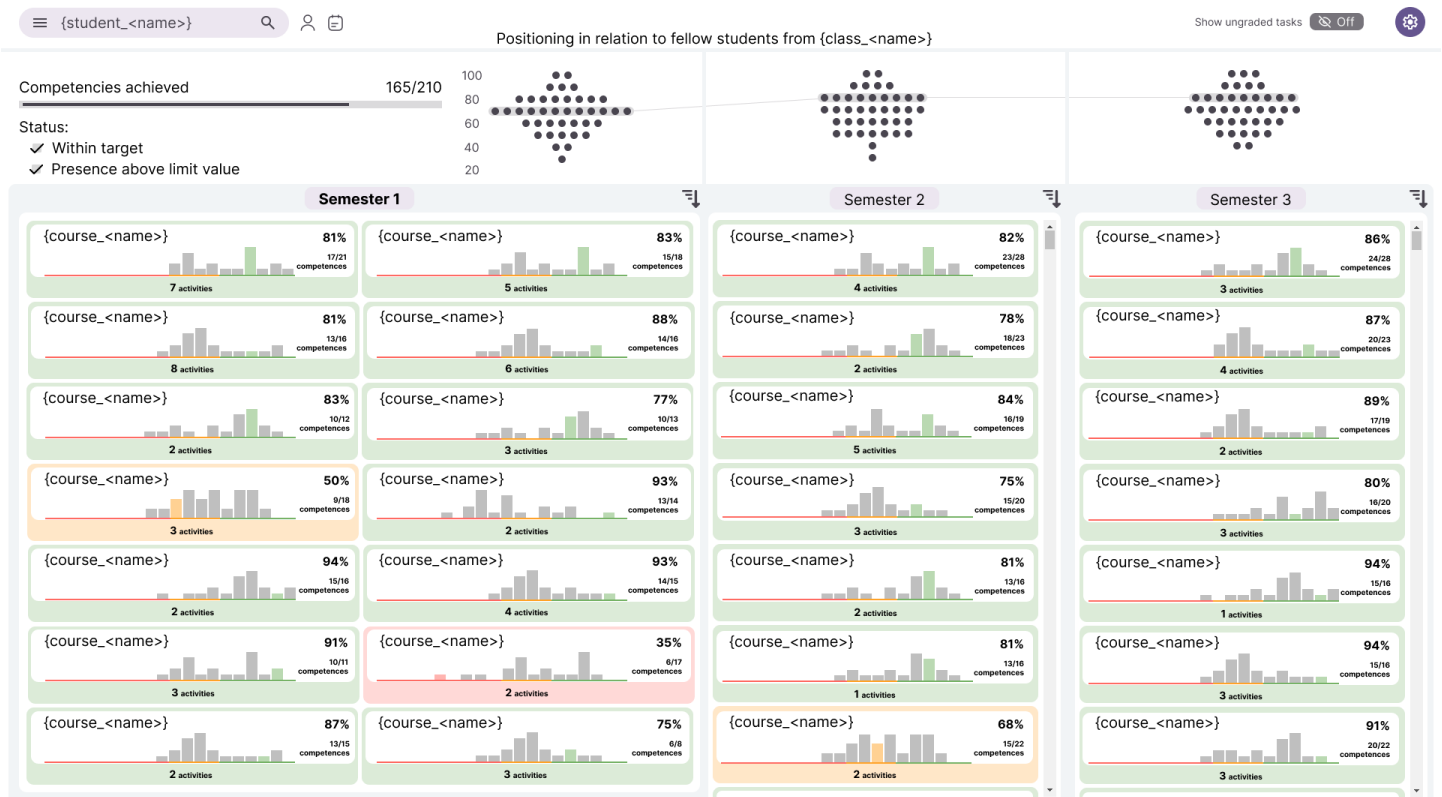


Key Features

- Student Selection & Notes – Filter students and access individual profiles and notes
- Competency Overview – Track overall status and achievement across semesters
- Subject Breakdown – Explore competency coverage, evaluation quantity, and class performance
- Competency Details – Dive into which specific competencies need more attention
- Activity Insights – Visualize student performance on evaluations vs. class metrics

Summary of the Dashboard Prototype

1. Student Overview Panel



This section allows the teacher to:

- Filter and select a student
- Open the student's profile
- Read and add teacher notes
- Toggle visibility of unevaluated tests/assignments
- Access general student information (e.g., number of achieved competencies, attendance, overall status such as "on track")

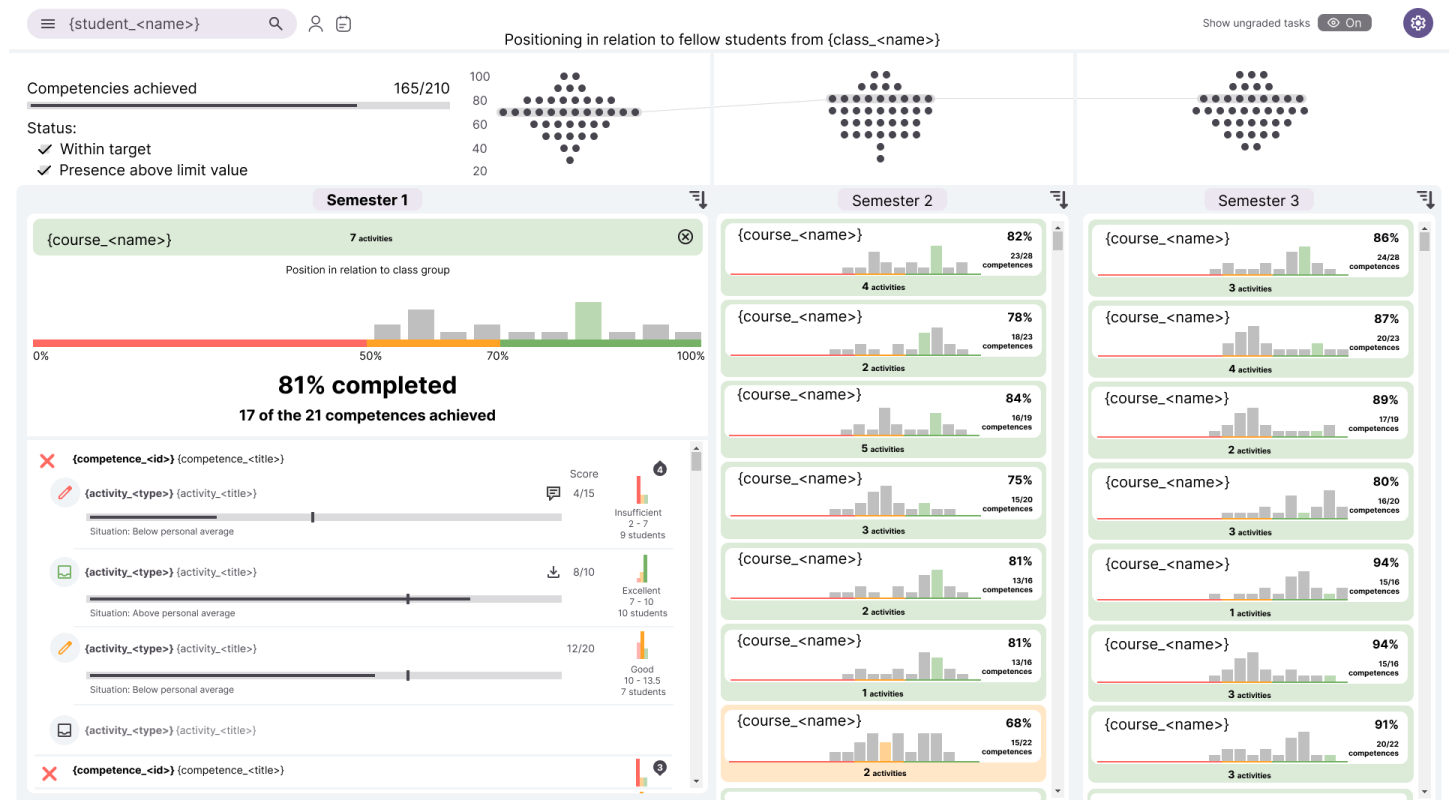
2. Semester Comparison & Subject Overview



This section provides:

- A three-semester comparison of the student's standing relative to classmates
- Visualizations per subject with:
 - Achieved competencies
 - Number of evaluations
 - Graphical class distribution
- A subject is clickable, leading to a detailed competency breakdown

3. Competency Details & Evaluation Breakdown



This deeper view shows:

- Which competencies are completed, nearly achieved, or need attention
- A breakdown of all tests/assignments contributing to the competency
- Per evaluation:
 - Student's progress visualization
 - Raw score
 - Class average comparison
 - Toggle to show/hide unevaluated items

API Mapping by Dashboard Area

Get Class Name



🔍 Zoek een leerling...

Competenties behaald

706/840



Marc Vertongen

Klas: 3STEM

Status: ⌚ ✓ Binnen doelstelling

Student Filter Logic



UI element: Student selector in top-left corner

Action: When selecting a student, CLAIRE fetches:

- Available classes (GET /api/classes)
- Students in those classes (GET /api/classes/{class_id}/students)
- Mapping between student and class
- Metadata for student previews (e.g., name, ID, status)

Zoek een leerling...

1Moderne 2Moderne 3STEM 3TW 3LAT 4Sport

4Mechanica 4Handel 5WeWi 6WeWi At Risk

Afwezigheidsrisico

1Moderne

Liam Janssens Mila Claes

Olivia Peeters Finn Dubois

Noah Maes Nora Vermeulen

Emma Jacobs Levi De Vos

Lucas Willems Ella Wouters

2Moderne

Adam Verstraeten Wout Mertens

Fien Vandenberghe Fleur Vandecasteele

Roos Hendrickx Seppe Verschueren

Stan Goossens Noor Vandermeulen

Lotte Verhoeven Tuur De Ridder

3STEM

Marc Vertongen Lotte Jacobs

Emma Peeters Seppe Claes

Thomas Janssens Fien Martens

Sophie Maes Wout Willems

On Student Selection

UI element: Clicking a student name

Action: The dashboard loads:

- Student-specific data (GET /api/students/{student_id})
 - Achieved competencies
 - Attendance
 - Semester comparison
- Class-level data for context (GET /api/classes/{class_id}/analytics)
 - Class averages, medians
 - Class distribution
 - Relative positioning per competency and subject

Subject Element (Course Overview)



Each course card in the subject overview section presents a compact but informative visualization of a student's progress within a specific subject ({course_name}).



Displayed Elements

- **Course Name** – The subject's title (e.g., "Mathematics", shown abstract as {course_name}).
- **Competency Score** – Displayed as a fraction, e.g., 17/21 competencies achieved.
- **Class Distribution** – A set of grey bars represents the distribution of students across performance groups.
- **Student Highlight** – The student's own group is visually highlighted (green = strong, orange = average, red = weak).
- **Border Color** – The border of the course card reflects the student's performance category.
- **Activity Count** – The number of registered activities (tests, assignments) contributing to the score is shown at the bottom of the card.

API Involved

- GET /api/students/{student_id}/courses
→ Retrieves all subjects linked to the student.
- GET /api/students/{student_id}/courses/{course_id}/competencies
→ Returns the number of achieved vs. total competencies for the course.
- GET /api/classes/{class_id}/courses/{course_id}/distribution
→ Provides performance distribution of the class for this subject and the student's assigned group.
- GET /api/students/{student_id}/courses/{course_id}/evaluations
→ Returns the list and count of activities that contribute to the course score.